

**ECON 101—Introduction to Microeconomics
Fall 2019**

Midterm II

November 11, 2019

Last Name:
First Name:
NU ID number:

Instructions:

- Answer all questions.
- Record your multiple choice answers on this page.
- For the open problems you will have space after each question to develop your answer.

Part I: Multiple choice

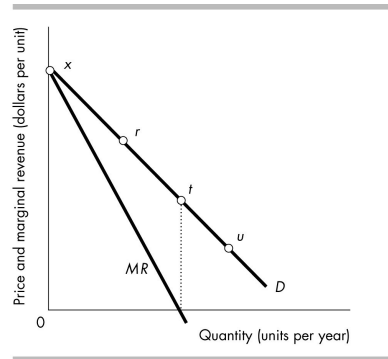
Please record your answers here.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

MCQ	P1	P2	Total
For instructor's use only. Do not write anything in here.			

Part I: Multiple Choice

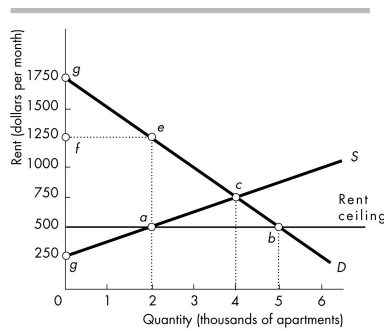
Please record all questions in the cover sheet. Each question is worth 4 points for a total of 60 points.



1. The figure above shows a monopoly firm’s demand curve. If the price and quantity of haircuts move from point *t* to point *u*, the monopoly’s _____
- (a) total revenue will fall.
 - (b) total revenue will rise.
 - (c) marginal revenue will increase.
 - (d) total revenue will remain the same.
2. Suppose a market has only one seller and only one buyer of a good in the market. The buyer is willing to pay \$50 for the good and the seller is willing to accept \$15. The market price of the good is determined to be \$30. If they trade, the social surplus will be _____.
- (a) \$45
 - (b) \$65
 - (c) \$15
 - (d) \$35

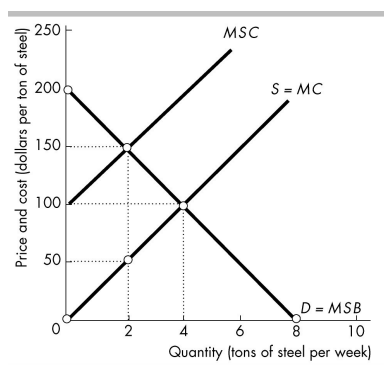
Quantity	Pat’s marginal benefit	Al’s marginal benefit	Marginal cost
1	10	4	0.5
2	7	3	1
3	4	2	1.5
4	2	0	2
5	1	0	2.5
6	0	0	3

3. If the good in the table above is a public good and Pat and Al are the only members of society, then the efficient quantity to produce is
- (a) 2 units.
 - (b) 4 units.
 - (c) 5 units.
 - (d) 3 units.



4. The above figure shows the demand and supply curves for housing. As a result of a rent ceiling at \$500, the deadweight loss is represented by the area

- (a) triangle acb
- (b) rectangle $feag$
- (c) triangle eca
- (d) triangle gfe



5. The above figure shows the marginal social benefit, marginal private cost and marginal social cost of producing steel. There is a marginal external _____ of _____ per ton.

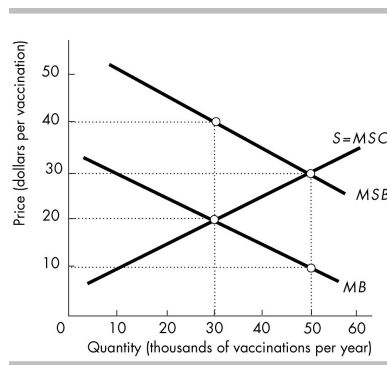
- (a) cost; \$200
- (b) benefit; \$200
- (c) benefit; \$100
- (d) cost; \$100

6. The above figure shows the marginal social benefit, marginal private cost and marginal social cost of producing steel. If the market is competitive and unregulated, the equilibrium quantity of steel is _____ tons and the efficient quantity of steel is _____

- (a) 2; 2
- (b) 2; 4
- (c) None of the above answers is correct.
- (d) 4; 2

7. \$20 is to be divided among two individuals—Gary and Jamie. Which of the following allocations is NOT Pareto efficient?

- (a) Gary receives \$15, and Jamie receives \$5.
- (b) Gary receives \$19, and Jamie receives \$1.
- (c) Gary receives \$8, and Jamie receives \$9.
- (d) Gary receives \$1, and Jamie receives \$19.



8. In the above figure, to achieve efficiency the government could offer a subsidy to producers of _____ per vaccination so that consumers would pay _____ per vaccination.

- (a) \$20; \$20
- (b) \$20; \$10
- (c) \$30; \$10
- (d) \$30; \$40

9. Which of the following can create a monopoly?

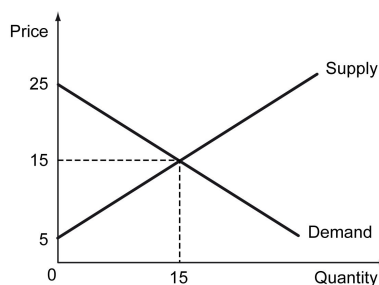
- I. high prices
- II. public franchise
- III. patent
- IV. government license

- (a) I and III
- (b) II, III and IV
- (c) I and II
- (d) I, II and III

10. The incidence of sales tax is determined by the

- (a) price elasticities of supply and demand.
- (b) federal government in all cases.
- (c) greed of the sellers.
- (d) level of government (for example, local, state, or federal) which imposes the tax.

The following figure illustrates the demand and supply of decorative light bulbs in a perfectly competitive market.



11. Refer to the figure above. What is the social surplus if the market is in equilibrium?

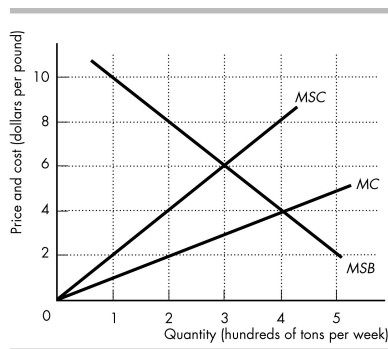
- (a) \$75
- (b) \$150
- (c) \$50
- (d) \$100

Quantity (millions of vaccinations)	Perceived marginal benefit	Marginal social benefit
1	12	18
2	11	16
3	10	14
4	9	12
5	8	10
6	7	8

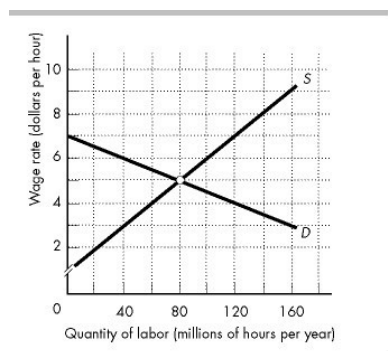
12. The table above gives information on the perceived marginal (private) benefit and marginal social benefit associated with vaccination against varicella (chicken pox). If the marginal cost of a varicella vaccination is \$10, to achieve an efficient quantity of vaccinations, the government could provide a subsidy of
- (a) \$1 per vaccination.
 - (b) \$2 per vaccination
 - (c) \$4 per vaccination.
 - (d) \$3 per vaccination.

Price (dollars per movie)	Quantity demanded, weekend (movies per week)	Quantity demanded, weekday (movies per week)
18	0	0
15	100	0
12	200	0
9	300	100
6	400	200
3	500	300

13. Roxie's Movie Theatre has a monopoly and discovers that at \$12 a movie, no one is buying movie tickets during weekdays. Roxie's conducts a survey and the table above reveals the results of the survey. Roxie decides to price discriminate between weekend and weekday moviegoers. The marginal cost of a showing a movie is \$6. Roxie's charges _____ on weekdays and _____ on weekends.
- (a) \$3; \$12
 - (b) \$6; \$18
 - (c) \$9; \$12
 - (d) \$6; \$15



14. The above figure shows the marginal private cost curve, marginal social cost curve, and marginal social benefit curve for cod, a common resource. A quota to prevent the overuse of the cod sets the catch equal to
- (a) 0 tons per week.
 - (b) None of the above answers is correct.
 - (c) 400 tons per week.
 - (d) 300 tons per week.



15. The figure above shows the market for low-skilled labor in Midland city in a nation to the south. The government sets a minimum wage at \$6 per hour. The minimum wage _____ the employers' surplus by _____ million per year.
- (a) decreases; \$120
 - (b) increases; \$100
 - (c) increases; \$40
 - (d) decreases; \$60

End of Multiple Choice Section: Please don't forget to record your answers in the cover sheet!

Continue to Part 2

Part II: Open problems

Instructions: Please do your calculations in the space provided. I will take into consideration your procedure. Please record your final answer in the space provided. There are two problems each worth 20 points.

1. (20 points) Suppose that AMD has a monopoly in the market for computer chips. AMD's total costs are given by $TC(X) = \frac{5}{2}X^2 + 15,000$. Thus, AMD's marginal cost is $MC(X) = 5X$.
 - (a) (10 points) The demand for computer chips is $X_D = 100 - 0.1P$. Find the level of output that maximizes Intel's profits. What price is Intel charging? What are its profits?

AMD is producing _____ computer chips, charging a price of _____ per chip and its profits are _____.

- (b) (10 points) What level of output would maximize total surplus in the computer chip market? Calculate the dead-weight loss generated by AMD behaving as a monopolist.

The efficient level of output is _____ and the dead-weight loss is _____.

2. (20 points) The demand for flu shots this season is given by $P = 13 - 0.0005Q$. The marginal cost of a flu shot is given by $MC = 8 + 0.0005Q$.

(a) (5 points) In a competitive market, what are the equilibrium price and quantity of flu shots?

The competitive equilibrium price is _____ and the equilibrium quantity is _____.

- (b) (10 points) Suppose that the *marginal external benefit* of flu shot is $MEB = 0.00025Q$. Compute the socially efficient quantity of flu shots.

The socially efficient quantity of flu shots is _____.

- (c) (5 points) Suppose that the government decides to give a subsidy of S per flu shot. What subsidy S should the government set to achieve the social optimum?

The per-flu shot subsidy that the government needs to put is _____.

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